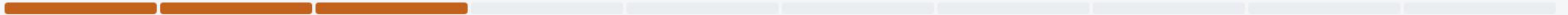


DGLD DOMAIN-GATED LATENT DIFFUSION

The high explosives we rely on are decades old

So we taught a diffusion model to design new ones, and let first-principles physics decide which are real.



Energetic materials

WHY Explosives and propellants run mining, demolition, rocket launch and car airbags, yet today's best compounds were designed decades ago.

- **What we optimise:** crystal density ρ , detonation velocity D and pressure P , alongside stability and cost.
- **The field has stalled:** the two workhorses were developed decades ago, **HMX** in the 1940s and **CL-20** in 1987, and few new HMX-class compounds have been disclosed since.
- **Why it is hard:** the design space is astronomically large and the goals fight each other, a molecule has to be new, dense, stable and actually makeable all at once.

HMX = “High-Melting eXplosive” (octogen), first made in the 1940s as a by-product of making RDX. **CL-20** = “China Lake compound 20”, synthesised there in 1987. The names are lab shorthand, not chemistry.

The goal: prescreen candidates computationally, so a lab only ever touches molecules already vetted on every axis.

Objectives: only one quadrant is a result

WHY A useful result is not simply high performance. It has to be high-performance **AND** genuinely new **AND** makeable.

KNOWN-GOOD

A lookup, not a discovery. HMX, RDX, CL-20 live here.

PRODUCTIVE

Novel and HMX-class, the only quadrant that counts.

NEITHER

Low performance, not novel.

JUNK

Novel but useless.

Vertical axis: predicted performance. Horizontal axis: novelty, the structural distance from everything already known.

DENSITY ρ

VELOCITY D

PRESSURE P

$\geq 1.85 \text{ g cm}^{-3}$

$\geq 9.0 \text{ km s}^{-1}$

$\geq 35 \text{ GPa}$

Three tests, together: **performance** in the HMX class, **genuinely new**, and **actually makeable**.

Generative models for material discovery

WHY DGLD is the same idea as the generative models you already know, moved into molecule space.

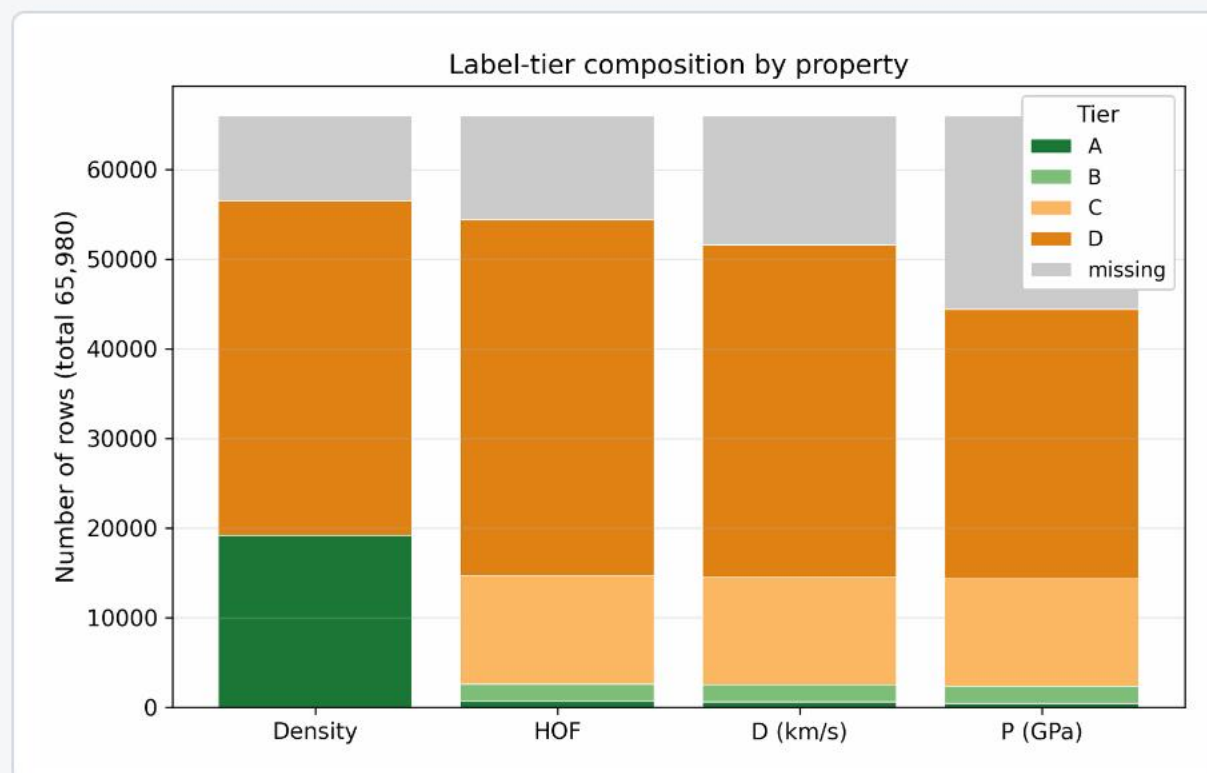
LLM	A prompt produces text . Pretrain on a large pile of unlabelled text, then fine-tune on prompt / response pairs.
Stable Diffusion	A prompt produces an image . Pretrain on many images, then fine-tune on image / text pairs.
DGLD (this work)	A target property vector (ρ , D, P, HOF) produces a molecule , written as SELFIES, a text encoding of chemical structure. Pretrain on ~694,000 mostly-unlabelled molecules, not all energetic, then fine-tune on known energetic molecules.

Same recipe every time: pretrain broad, fine-tune narrow, steer at generation.
DGLD applies it to molecules instead of words or pixels.

Training-data labels are very sparse

WHY The property labels we train on are uneven and mostly low-trust, and the model has to be built with that in mind.

TIER	WHERE THE NUMBER CAME FROM	ROWS	TRUST
A	Measured in a laboratory	~19,200	yes
B	Computed from first principles (DFT)	~1,900	yes
C	A 1968 closed-form formula (Kamlet–Jacobs)	~12,000	no
D	Machine-learning surrogate (Uni-Mol)	gap-fill	no



Label-tier composition per property (total 65,980 rows). Only density is mostly measured; velocity and pressure are mostly formula or surrogate.

Density is measured, by diffraction. Detonation velocity is not: measuring it means making and firing the compound. So most D and P labels are computed, and the model is trained with those limitations in mind.

Beyond performance: novelty and synthesis

WHY A molecule that scores well on physics is still worthless if it is already known or impossible to make.

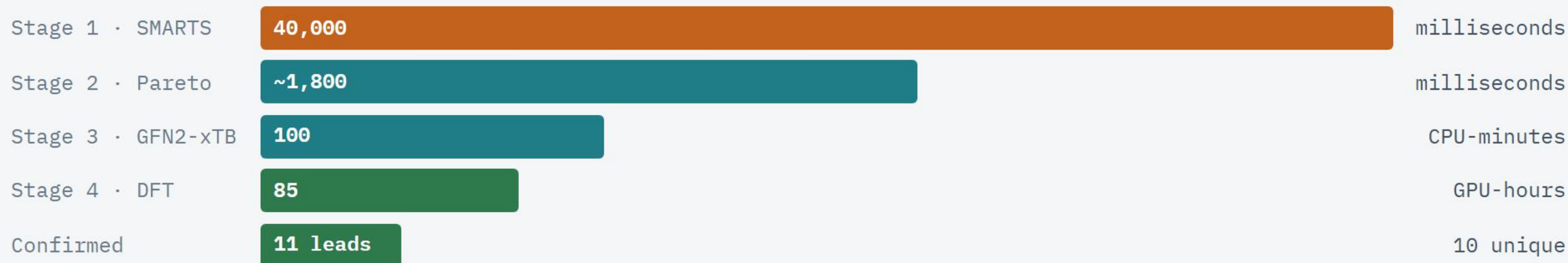
Novelty is checked against three references:

- **PubChem**, the public catalogue of known compounds.
- **SureChEMBL**, structures extracted from patents.
- The full **694,000-molecule** training pool.

SCORE	HOW IT WORKS	WE REQUIRE
Tanimoto similarity	Chop both molecules into fragments; what fraction do they share? 1.0 identical, 0 nothing in common.	≤ 0.55 vs every training molecule best lead 0.27
SA synthetic accessibility	Scores 1 to 10 by how rare the fragments are: common low, exotic high.	≤ 5.0
SCScore synthetic complexity	Scored 1 to 5, learned from millions of published reactions: roughly the distance from purchasable starting materials.	≤ 3.5

Cheap filters on everything, expensive physics on survivors

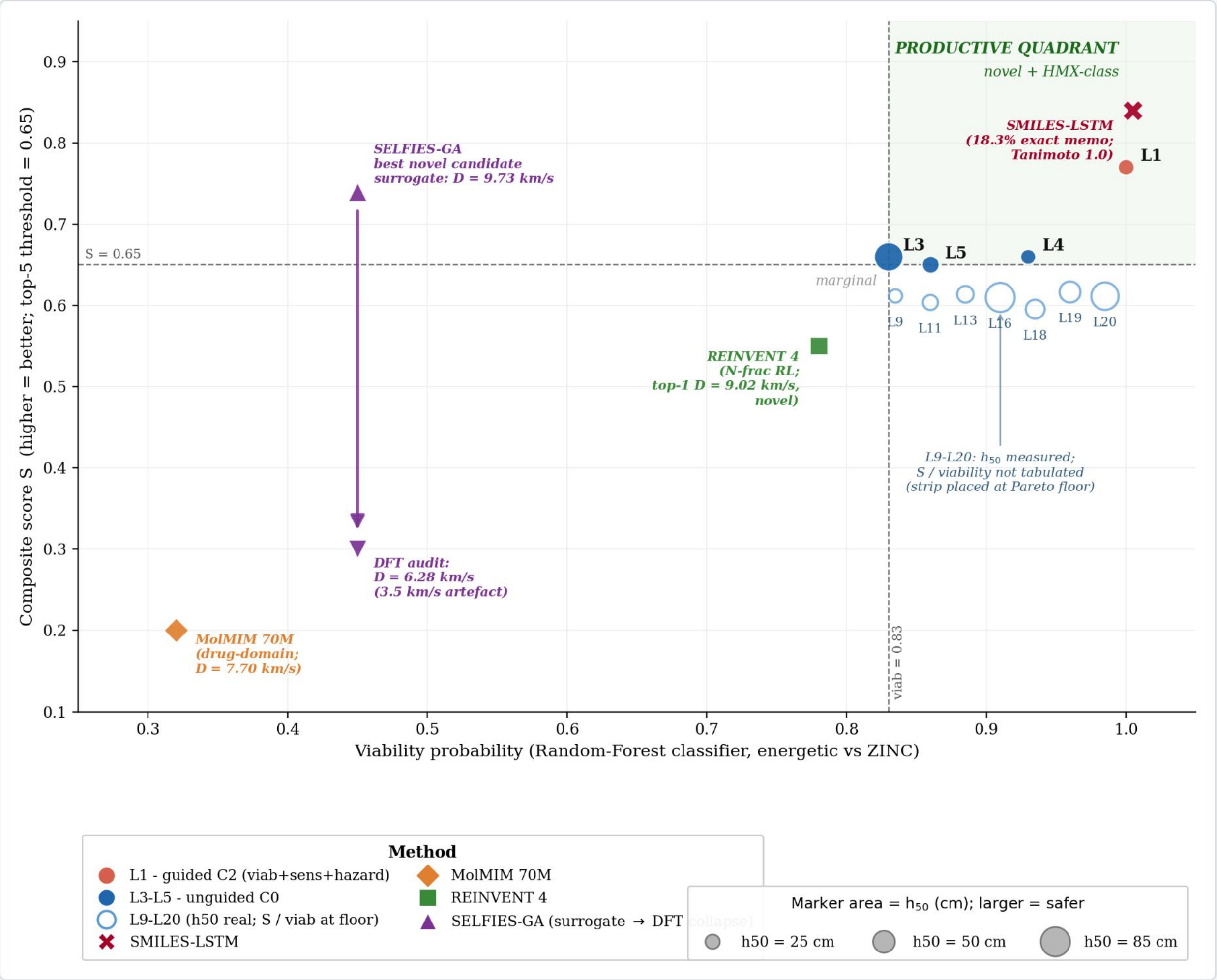
WHY The generator is cheap and real physics is not, so every candidate meets the cheap filters first and only survivors reach quantum chemistry.



- **SMARTS**, a standard substructure pattern language (the kind RDKit reads). The reject rules are two sources: **hand-curated by us** (radicals, sulphur, halogens, bad valences) layered on the published **Bruns–Watson** structural-alert list, itself distilled from ChemAxon “dangerous reactivity” patterns.
- **Pareto front**, the set of candidates nothing else beats on *every* objective at once.
- **GFN2-xTB**, fast semi-empirical quantum chemistry, roughly a thousand times cheaper than DFT.
- **DFT**, accurate first-principles quantum chemistry, but hundreds to thousands of times slower than xTB.

Position against every baseline

WHY Plotted against every competing method, DGLD is the only one whose novel candidates stay on-target under real physics.



Every method placed by **viability** (x) and **composite score S** (y); marker size is impact insensitivity h_{50} (larger = safer). The green **productive quadrant** (top-right) is high-viability and high-composite. DGLD's leads sit there; SMILES-LSTM is a memorised copy (Tanimoto 1.0), MolMIM and REINVENT 4 fall short, and SELFIES-GA's best novel candidate collapses from surrogate D = 9.73 to DFT 6.28.

x · Viability Random-Forest probability that the molecule is a real energetic (trained: energetics vs drug-like ZINC). Right = more plausible.

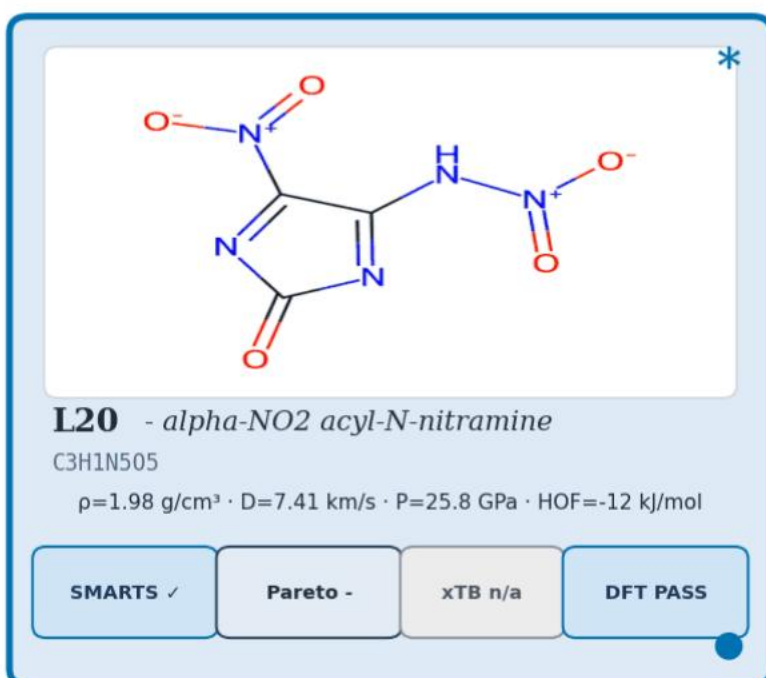
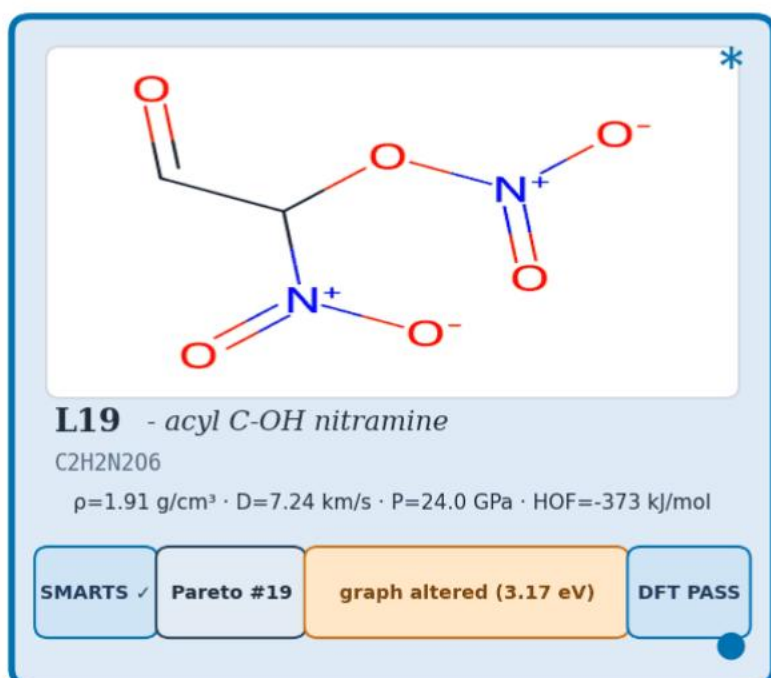
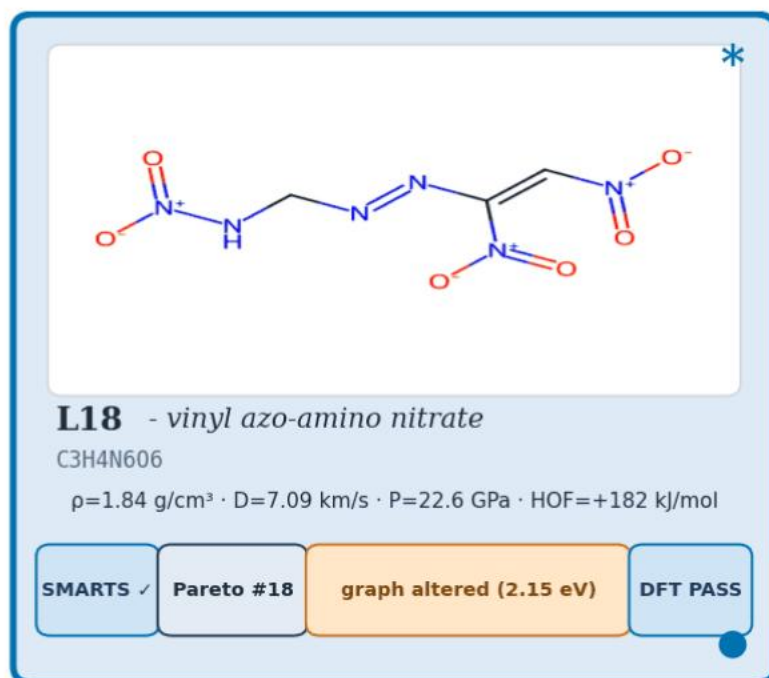
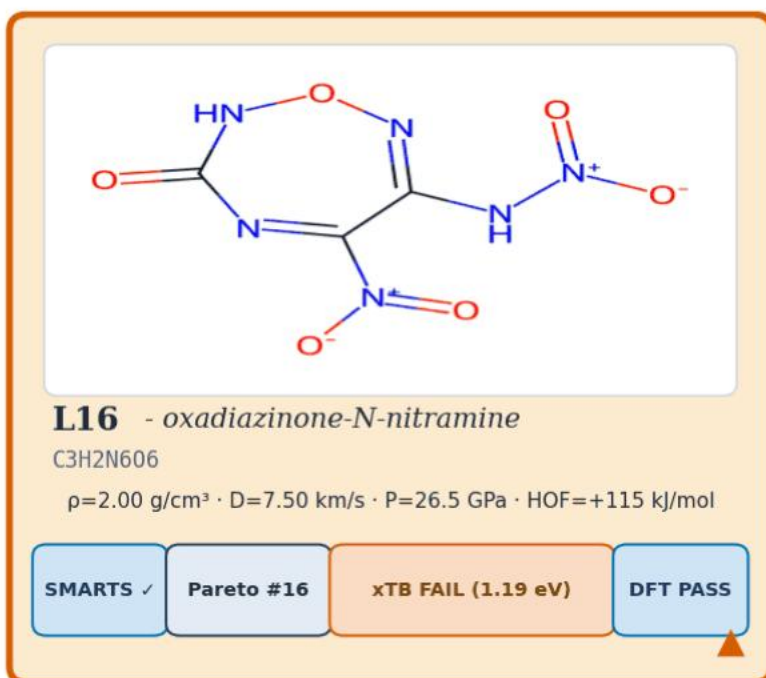
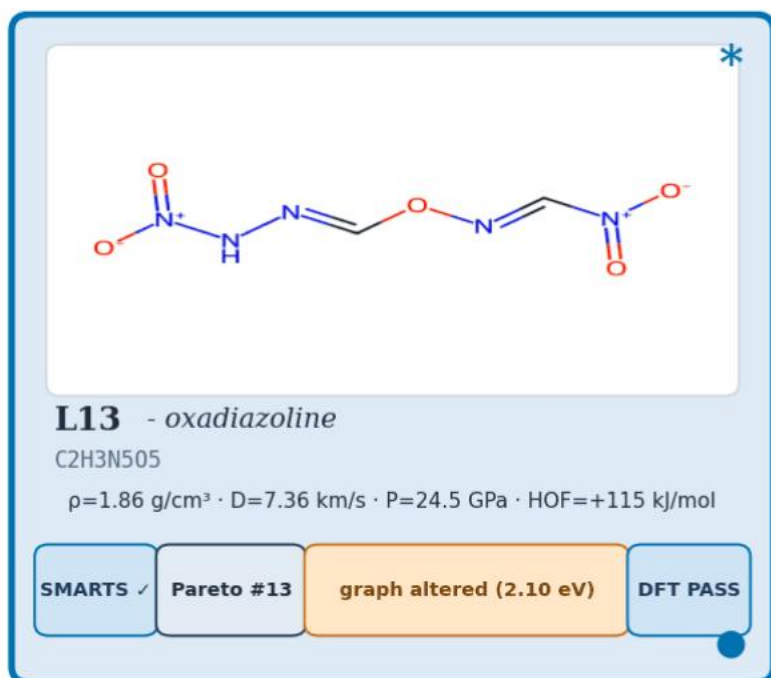
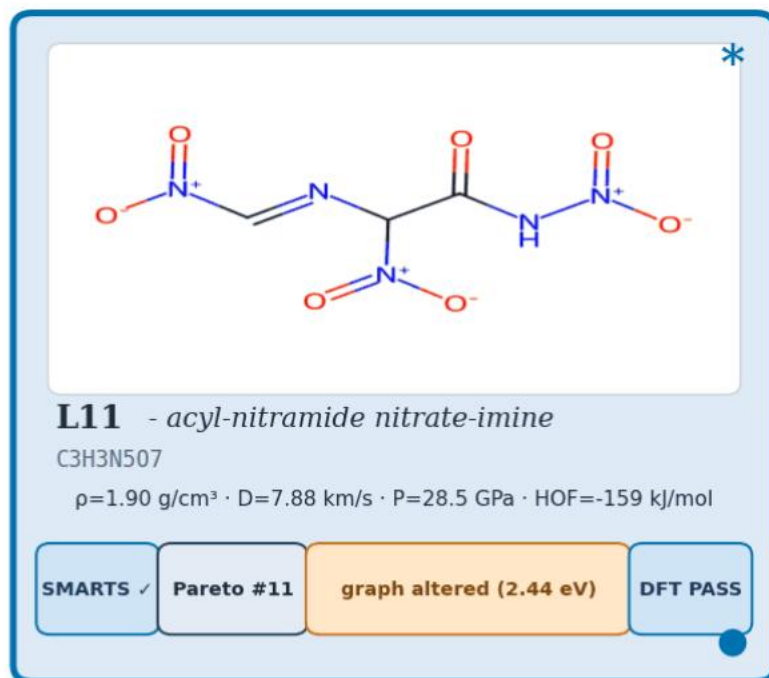
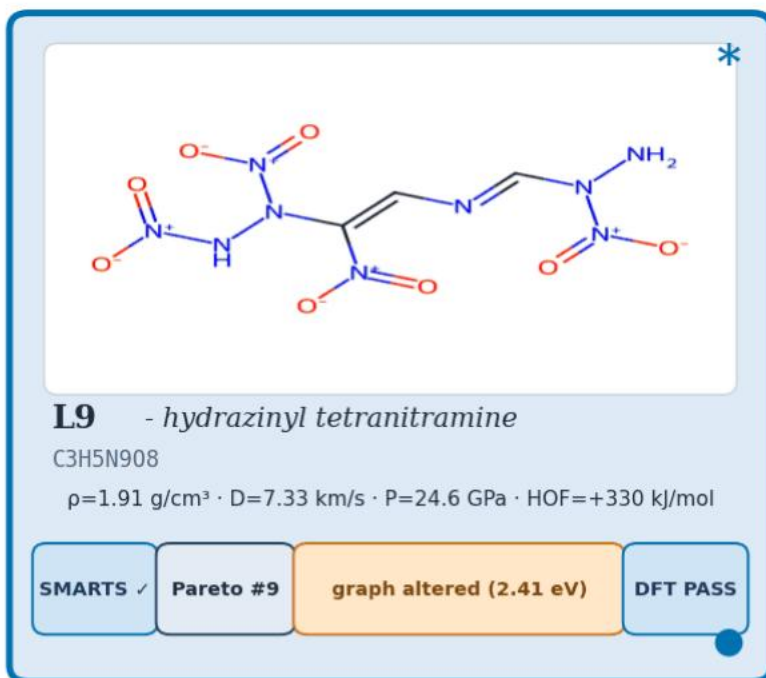
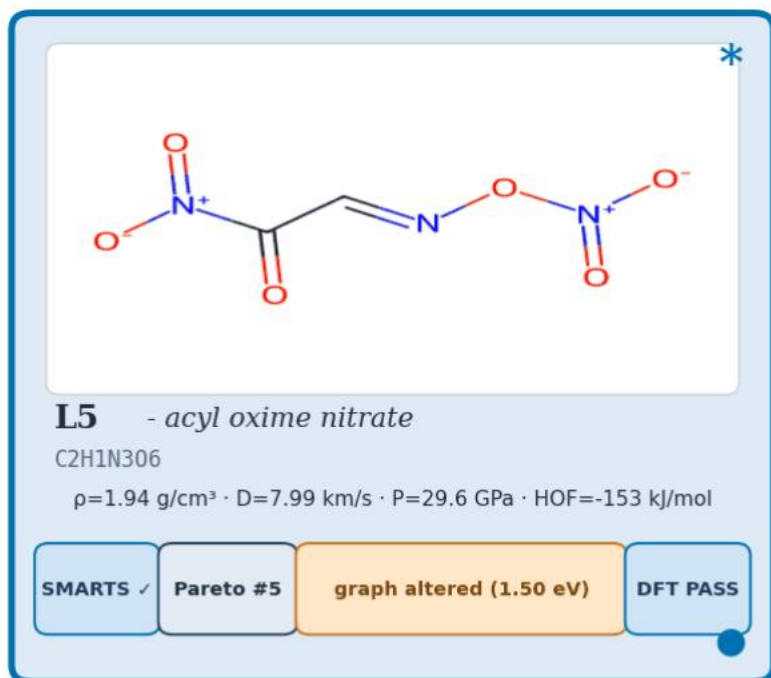
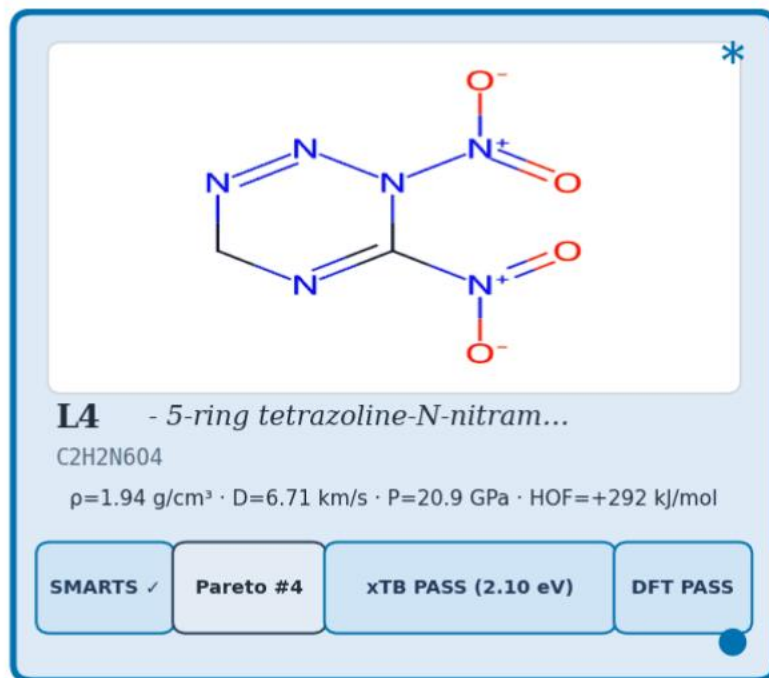
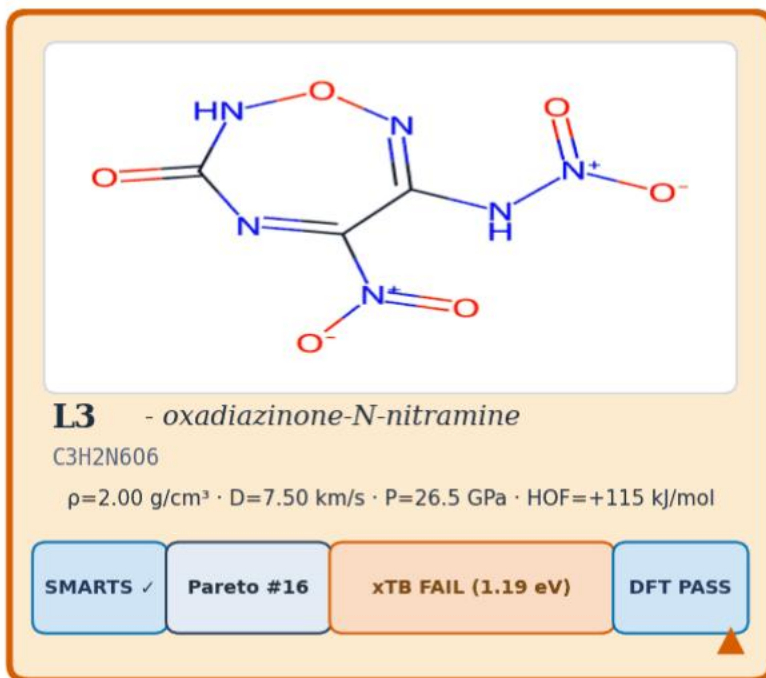
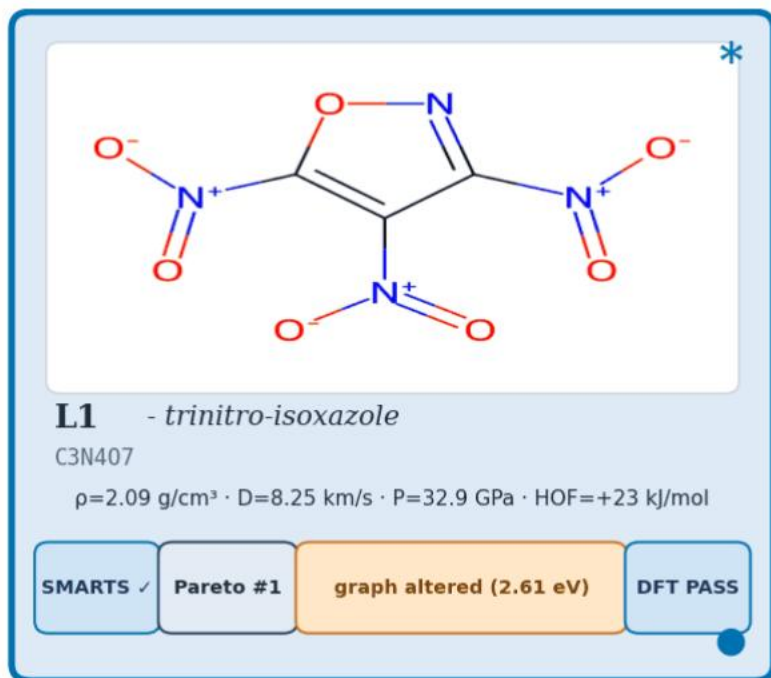
y · Composite score S One 0-to-1 number folding together all the objectives, performance, viability, novelty, sensitivity and hazard alerts, into a single rank. Higher = better; the top-5 band sits above 0.65.

dot size · h_{50} Impact sensitivity: drop a standard weight, h_{50} is the height at which half the drops detonate. A **bigger dot means a higher h_{50} , so a safer molecule**, harder to set off by accident.

Eleven chem-pass leads

WHY The survivors of the funnel, each with its structure, anchor-calibrated DFT properties, and validation trail.

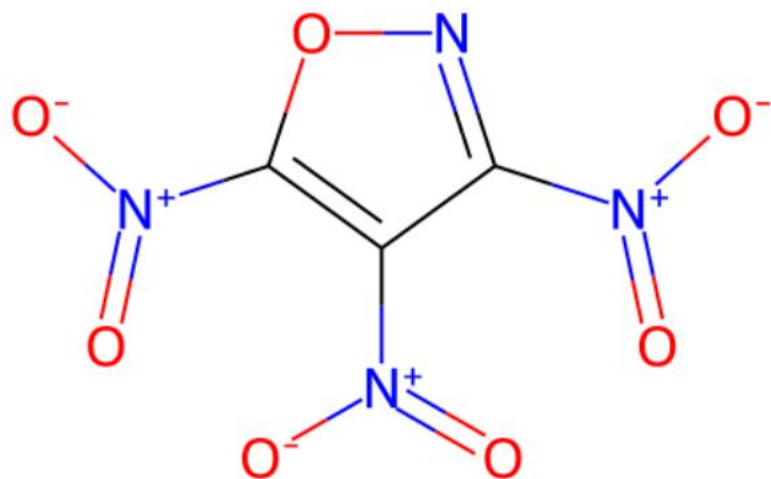
Eleven chem-pass DGLD leads: structure, anchor-calibrated DFT properties, and validation status



Eleven chem-pass DGLD leads (10 unique compounds). Each card carries the structure, ρ / D / P / HOF on the experimentally-anchored calibrated scale, and the SMARTS $\rightarrow$ Pareto $\rightarrow$ xTB $\rightarrow$ DFT status trail.

The candidate the model produced

WHY The headline lead, generated from noise, not retrieved from any database.



L1 · 3,4,5-trinitro-1,2-isoxazole

PROPERTY	VALUE	READ AGAINST
Density	2.09 g cm ⁻³	HMX 1.91
Detonation velocity	8.25 km s ⁻¹	HMX 7.52, PETN 8.24 (same scale)
Detonation pressure	32.9 GPa	
Similarity to nearest training molecule	0.27	1.0 would be a copy
Electronic stability gap	2.6 eV	1.5 eV is our pass mark
Weakest bond	86 kcal mol ⁻¹	below 80 would flag danger

Not in PubChem, not in the patent corpus, not in anything we trained on.

What counts as a good answer

WHY Before judging any method, be explicit about what a useful molecule has to satisfy. Four requirements, and they **PULL AGAINST EACH OTHER**.

1 · PERFORMANCE

Release a lot of energy, very fast. Measured as detonation velocity D and pressure P , both driven mostly by **crystal density**. Target: the HMX / PETN class, today's benchmark high explosives.

2 · STABILITY

Do **not** go off when dropped, struck or warmed. Measured as impact sensitivity h_{50} , plus thermal stability and the absence of known-dangerous groups.

3 · SYNTHESIS

Somebody has to be able to **make** it, from starting materials that exist, in a sane number of steps. A molecule nobody can build is a drawing.

4 · NOVELTY

It must be genuinely **new**: not in the chemical databases, not in the patent literature, not a re-run of the training set. Otherwise it is a lookup, not a discovery.

The difficulty is that these conflict. The usual way to raise energy release is to pack in more nitro groups (oxygen-rich $-\text{NO}_2$ units) and strain, which is also the usual way to make a molecule dangerously sensitive. Anyone can win on one axis alone. The search is for a molecule that holds **all four at once**, and every stage that follows exists to enforce one of them.

The data is uneven in three separate ways

The corpus is assembled from several public sources that differ in **both reliability and chemical scope**. Nothing about it is uniform, and each kind of unevenness needs its own answer.

1 · CHEMISTRY

Not everything is an explosive, and that is deliberate. Broad chemistry does three jobs: a checkpoint pretrained on general molecules teaches the encoder what a molecule even is; a ~380 k unlabelled CHNO (carbon, hydrogen, nitrogen and oxygen only) dump teaches the prior what plausible chemistry looks like; drug-like molecules act as negatives.

2 · COVERAGE

Few rows have all four properties. One molecule may have a measured density but no heat of formation. Each row carries its own list of which properties are actually present.

3 · PROVENANCE

A number is not just a number. Some were measured, some computed, some produced by a formula, some guessed by a model. They are not interchangeable.

TOTAL MOLECULES SEEN IN TRAINING

~694,000

66k labelled + 380k unlabelled CHNO + motif-augmented (grafted-on energetic groups), after dedup

MOLECULES WITH REPORTED PROPERTIES

~66,000

of those, the ones carrying detonation numbers

WHOSE NUMBERS YOU CAN TRUST

~3,000

measured in a lab, or computed with quantum mechanics

The other 95 % are estimates, and **the estimates are not equally good either**. A model trained on all of it imitates the estimates, and proposes molecules that collapse once real physics is applied.

Sort every label by how much you trust it

Unevenness 3 gets the architectural answer: **every individual label is tagged with how it was produced**. Unevenness 2 is handled by the same mechanism, since a row simply carries no tag for a property it lacks.

TIER A

Experimental

Somebody made the compound and measured it in a lab.

TIER B

DFT

Computed from quantum mechanics: no lab, but no guessing either.

TIER C

Kamlet–Jacobs

A 1968 formula. Instant, but unreliable outside the chemistry it was fitted on.

TIER D

ML surrogate

Predicted by the Uni-Mol 3D ensemble. A guess about a guess.

Tier C is the Kamlet–Jacobs **formula**, a fixed equation with no learned parameters. Tier D is **Uni-Mol**, a **neural network**: pretrained on millions of molecules in **3D**, so it judges a candidate by its shape in space rather than by its formula. Ours is an in-house ensemble fine-tuned on detonation data (~84 M parameters per fold, two folds). It answers in milliseconds where DFT takes an hour, which is why it can label the whole corpus, and why its answers are the least trusted tier.

Uni-Mol: Zhou, G. et al. (2023), *Uni-Mol: A Universal 3D Molecular Representation Learning Framework*, ICLR 2023. Kamlet & Jacobs (1968), *J. Chem. Phys.* **48**, 23–35.

The gate is graded, not a switch. Every label carries a trust weight, Tier A 1.0, B 0.7, C 0.3, D 0.1. Trusted labels steer far more often; Tier C/D are **down-weighted**, **not silenced**, and they also shape the unconditional prior. Nothing is discarded, low-trust labels just get a quieter vote.

We do not train Uni-Mol from scratch. We adapt it.

WHY Pretraining a 3D molecular model needs millions of molecules and a budget we do not have. Somebody already did it, on general chemistry. We only have to teach it **OUR** question.

- 1** **Take** the public Uni-Mol checkpoint, pretrained on millions of molecules in 3D (Zhou et al., ICLR 2023). Not ours, not energetic-specific. existing model
- 2** **Build a 3D shape** for each molecule first: the text has no coordinates, so RDKit generates a conformer (ETKDGv3) and relaxes it with the MMFF94 force field This is why Uni-Mol is a label-maker, not a per-step guide: a molecule is built over 40 denoising steps, and scoring it inside the loop would mean decoding and rebuilding a 3D conformer at all 40, ruinous in cost and impossible to push a gradient through. conformer
- 3** **Fine-tune** it on our labelled energetic data, as a regression model over eight outputs ρ , D, P, T, E, V, HOF, BDE our data
- 4** **Ensemble two folds:** split the labelled data by **scaffold** so each model trains on different core skeletons, fine-tune both from the same checkpoint, then average their two predictions scaffold-split, not random, because we generate novel scaffolds; the gap between the folds is a cheap uncertainty signal. Both are fine-tuned, neither is the raw checkpoint. $2 \times \sim 84 \text{ M}$

WHY TWO FOLDS, AND WHAT "SCAFFOLD" MEANS

A **scaffold** is a molecule's bare core skeleton. We use the **Bemis–Murcko scaffold** exactly: the ring systems plus the linkers joining them, with every side chain stripped off. Molecules are grouped by this scaffold, and the groups are split into **two disjoint folds**, so no Bemis–Murcko scaffold appears in both. Fold 1 learns on one set of cores, fold 2 on another.

Why bother? DGLD invents *new* scaffolds, so each fold is scored on cores it never trained on, the real deployment test. A random split would let both folds memorise the same cores and look better than they are on genuinely new chemistry.

Trained on tiers A + B + C only. That is deliberate: it learns from experimental, DFT and formula-derived rows, then *produces* the Tier D values for molecules that have nothing. Training it on its own output would be circular. Held-out quality is $R^2 = 0.84$ to 0.92 in 5-fold cross-validation, good enough to rank candidates and not good enough to trust as a final answer, which is precisely why Tier D sits at the bottom and why the DFT stage exists.

Two different things are called “labels”

WHY The word is overloaded in this field, and mixing the two up makes the whole architecture look arbitrary. They come from different places and do different jobs.

PROPERTY TARGETS

The four detonation numbers: ρ , D , P , HOF (density, velocity, pressure, heat of formation).

Where from: the dataset. Measured, computed or estimated, which is exactly what the trust tiers rank.

Used for: conditioning the denoiser (the network that generates molecules). They are the “what to aim for”, gated by the mask, a per-molecule on/off switch.

Coverage: patchy. Many molecules are missing some.

QUALITY SIGNALS

Viability, sensitivity, hazard (plus SA and SC, two drug-domain scores for how hard a molecule looks to make).

Where from: nowhere. Nobody reports them, so we **manufacture** them for every molecule from rules, correlations and a classifier.

Used for: training the score model (the judge that can read a half-finished molecule), which then steers sampling. Never conditioning.

Coverage: complete, by construction.

Keep the split in mind for the rest of the deck: **property targets are conditioned on; quality signals are steered by**. The mask governs the first group only, which is why it has exactly four bits.

How DFT and Kamlet–Jacobs actually work

Tiers B and C are both computed rather than measured, but they are not the same kind of computation, and they run in sequence.

DFT

STRUCTURE → ρ , HOF

Density functional theory. A quantum-mechanics calculation of a molecule's shape and energy from the behaviour of its electrons. **Takes only the structure** and returns optimised geometry and electronic energy, giving **heat of formation HOF**, with **density ρ** from a van-der-Waals volume on that geometry. The raw calculation uses no experimental input (so it is *first principles*), though every ρ , D and P we report is afterward put on a calibrated scale (below). About an hour of GPU time per molecule: accurate, far too slow for thousands of candidates.

Kamlet–Jacobs

ρ , HOF → D, P

A closed-form formula from 1968. Takes the density and heat of formation from DFT plus the atom counts, and returns **detonation velocity D** and **pressure P** instantly. Being an empirical fit, it drifts outside the chemistry it was built on, notably for nitrogen-rich molecules, exactly the region we are searching.

structure → DFT → ρ , HOF → Kamlet–Jacobs → D, P → **6-anchor calibration**. Neither step is a direct measurement, so a final linear fit to the measured values of RDX, HMX, TATB, PETN, FOX-7 and NTO ties every quoted number back to experiment.

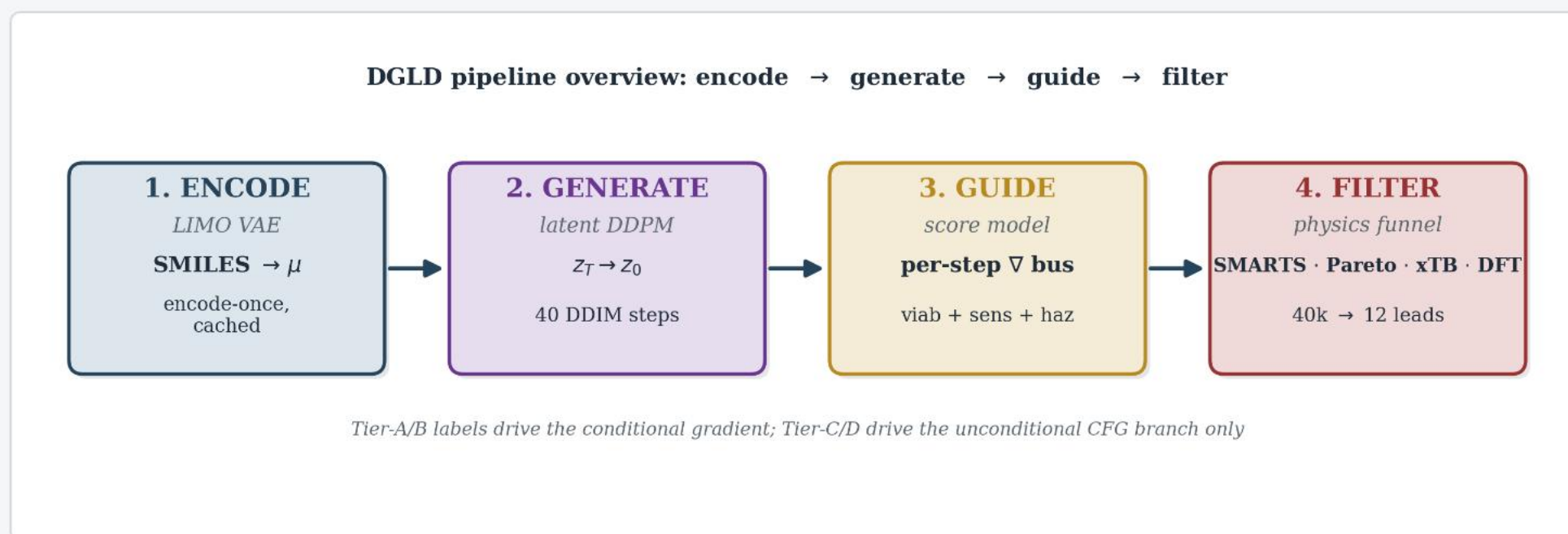
Why each stage exists

WHY Every stage exists because a specific obstacle stands between “we want a molecule” and “here is one worth making”. Nothing in the design is decorative.

THE OBSTACLE	WHAT IT FORCES	STAGE
Molecules are discrete text. Diffusion only works on numbers that vary smoothly.	Learn a numeric space where similar molecules sit close together.	Encoder
The properties we want to aim at are mostly untrustworthy estimates.	Rank each label by provenance, and let only reliable ones steer.	Trust tiers
Nobody publishes “is this viable” or “is this safe”.	Manufacture those signals for every molecule in the corpus.	Labels
Mid-generation there is no molecule yet, only a noisy vector. Ordinary chemistry tools cannot read it.	Train a judge that works on the half-formed state.	Score model
Something has to actually invent structures, not just rate them.	Train a generator, conditioned on the targets.	Denoiser
One run explores one region and stops.	Run several differently-tuned lanes and pool the results.	Sampling
A model's own score is an opinion. Surrogates can be gamed, and they were.	Re-test every survivor with real physics before reporting it.	Funnel

Four training stages, then sample, then filter

T1	Compressor: the LIMO VAE, a compress-and-rebuild network, turns molecules into vectors	~34 M, frozen
T2	Label pipelines: viability, sensitivity, hazard, performance	offline, once
T3	Score model: six heads that judge a noisy molecule	~8 M
T4	Denoiser: the generator, gated by the trust mask	2 × 44.6 M
S	Sampling: noise → 40 guided steps → decode → pool fusion (merge the lists)	≥40k / lane
F	Funnel: substructure rules (SMARTS) → Pareto → xTB → DFT	4 stages



Paper Figure 5. The same pipeline as printed in the manuscript.

Everything from T1 onward operates on 1024-number vectors, called **latents**, never on text.

Compress every molecule into a vector

WHY Diffusion works by nudging numbers, so it needs a space where **SIMILAR MOLECULES SIT CLOSE TOGETHER**. Chemical text has no usable sense of “nearby”, so we build that space first.

Molecules are written as **SELFIES** strings, a notation where every string is a valid molecule. A variational autoencoder (LIMO) learns to squeeze each one into **1024 numbers** and rebuild it.

INPUT

O=[N+]([O-])c1noc...

LATENT VECTOR

[0.42, -1.10, ...]

1024 dimensions

REBUILT

O=[N+]([O-])c1noc...

Trained once, then **frozen**, and every latent is cached. From here on the pipeline is pure vector maths, which is what makes diffusion practical on a small GPU budget.

“Energetic-biased” means the corpus is deliberately skewed toward explosive chemistry, not general molecules. It is built from three pools: the 66k labelled master, a ~380k unlabelled dump of **CHNO** molecules (built only from carbon, hydrogen, nitrogen and oxygen, the elements explosives are made of) that carry no property numbers, and a large **motif-augmented** expansion that programmatically grafts nitro, nitramine, azide, cyano, tetrazole and oxadiazole groups onto known energetic scaffolds (core molecular skeletons). The encoder therefore spends its fine-tuning budget on the chemistry we care about instead of on drug-like molecules.

MLP-VAE, 1024-d latent, ~34 M parameters. Starts from the published LIMO checkpoint trained on general chemistry, then fine-tuned ~8,500 steps on 326 k energetic-biased SMILES, the standard text notation for a structure.

Where the extra million molecules came from

WHY 66,000 real molecules is not enough to teach a generative model the grammar of a whole chemical family. So we manufacture more, cheaply, by recombining what is already known.

THE SAME IDEA AS IMAGE AUGMENTATION

This is **data augmentation**, and a standard move. To train an image model you flip, crop and recolour each photo so it learns what a cat looks like from any angle, cheaply, without shooting new photos. Here we do the molecular version: recombining known parts, a textbook enumeration technique, so the model learns what energetic chemistry *looks like* without synthesising anything. **One real difference:** a flipped cat keeps its “cat” label, but our grafted molecules are brand new and carry **no** label, which is why they only shape the prior and never act as targets.

- 1** **Start from the scaffolds** of the labelled master: the bare core skeletons, mostly ring systems, left after stripping the decorations off known energetic molecules real cores
- 2** **Graft on energetic groups** by script: nitro, nitramine, azide, cyano, tetrazole, oxadiazole, in every position the valences allow a program, not a chemist
- 3** **Canonicalise and de-duplicate**, since many substitutions land on the same molecule ~1.08 M → ~230 k kept

These molecules are hypothetical. Nobody made them, nobody measured them, and we have no idea whether most are any good. That is precisely why they carry **no labels** and are used only to train the unconditional prior: they teach the model what energetic chemistry *looks like*, never what to aim for. Real knowledge still comes from the 66k labelled rows.

Three jobs, and only the last one touches generation

WHY To steer generation we need a **CLASSIFIER**, a model that scores how good a molecule is. Training a classifier needs labelled examples, and the labels we need do not exist, so we create them.

Which labels? The three **chemistry-quality** signals: **viability** (is this a real energetic molecule?), **sensitivity** (will it go off too easily?) and **hazard** (does it contain a known-dangerous group?). Nobody publishes these, so we derive one for every molecule from domain knowledge, physics correlations and rules. These are separate from the four property targets p/D/P/HOF, which are the trust-tiered dataset labels; here we mean the quality signals that steer.

- 1 **Derive** a label for every molecule in the corpus, **once, before training starts** offline
- 2 **Learn** a classifier that reproduces those labels from a *half-noised* molecule the score model
- 3 **Steer** generation with that learned model, once at every sampling step at sampling

The generators below never steer anything. They exist to create the training signal for step 2.

Build the four label pipelines

Four generators, three kinds of knowledge. None of them steers anything; they exist to create the training signal for step 2.

SIGNAL	QUESTION	HOW IT IS COMPUTED	KIND
Viability	Is this a real energetic molecule?	Random Forest (a vote of many decision trees) on Morgan fingerprints (substructure bit patterns) + RDKit descriptors (open-source toolkit numbers)	learned
Sensitivity	Will it go off too easily?	Politzer–Murray bond-strength fit on measured h_{50} data	measured physics
Hazard	Does it contain a known-dangerous group?	Our own energetic-specific SMARTS reject rules, plus the published Bruns–Watson demerit list (distilled from ChemAxon “dangerous reactivity” patterns)	expert rules
Performance	How strong would it be?	Uni-Mol ensemble → density, velocity, pressure, heat of formation	learned

Two different situations. Performance (ρ , D , P , HOF) is sometimes reported, so a row takes the measured value where one exists and a computed one otherwise: that is what the trust tiers rank. Viability, sensitivity and hazard are never reported, so they are derived for **every** row.

Uni-Mol ensemble ~84 M parameters per fold × 2 folds; the Random Forest, SMARTS catalogue and Politzer–Murray fit are not neural networks.

Tools that describe a molecule

Two of these describe a molecule *going in* to a classifier.

A fingerprint is not a label. A Morgan fingerprint only says *which substructures are present*; it has no opinion about whether the molecule is a good energetic material. The Random Forest is the function that turns those bits into a number. Fingerprint goes in, viability score comes out. You need both.

TWO DIFFERENT CLASSIFIERS, DO NOT CONFUSE THEM

The Random Forest here is a **label-maker (a teacher)**: it reads a finished molecule's fingerprint and computes a viability **label**, once, offline. It is **not** the classifier that steers generation. That is the **score model (the student)**, trained to reproduce these computed labels from a *half-noised latent*.

Why not just compute the label during generation? Mid-generation there is no finished molecule to feed the Random Forest, and steering needs a **differentiable** gradient it cannot give. So we compute the label once on real molecules, then train a differentiable student to predict it from the noisy latent, the only thing that can run in the sampling loop.

Morgan fingerprint

INPUT → VIABILITY

Represents: which substructures a molecule contains. Turns a molecule into a fixed list of 1s and 0s: every small circular patch of atoms and neighbours is hashed into a slot. **Molecules that share substructures light up the same slots**, which is what lets a classifier compare them.

RDKit descriptors

INPUT → VIABILITY

Represents: bulk make-up, size and composition. RDKit is the standard open-source chemistry toolkit; its descriptors are plain computed numbers such as **molecular weight, ring count, number of nitro groups, oxygen balance**, the last being whether a molecule carries enough oxygen to burn itself completely.

Tools that produce a value

The rest produce a value *coming out*: a number, or a yes/no class.

Politzer-Murray

PREDICTS → SENSITIVITY

Represents: how weak the weakest bond is. Published correlations linking **the strength of a molecule's weakest bond** to how easily it detonates on impact. A weaker trigger bond means a more sensitive explosive.

h₅₀

THE SENSITIVITY PROPERTY ITSELF

Represents: measured impact sensitivity, in centimetres. Drop a standard weight from increasing heights; **h₅₀ is the height in cm at which half the drops cause detonation.** Higher is safer: TATB, an unusually insensitive explosive, sits near 490 cm, primary explosives at a few cm.

SMARTS + Bruns-Watson

DECIDES → HAZARD

Represents: presence of a known-dangerous group. A yes/no class, not a number. SMARTS is a **pattern language for substructures**, regular expressions but for chemistry. Our hand-written catalogue is extended with the published **Bruns-Watson demerit list**, which penalises groups known to be reactive or unstable.

Uni-Mol ensemble

PREDICTS → PERFORMANCE

Represents: the four target numbers ρ , D, P and HOF. Uni-Mol is a pretrained model that reads molecules in **3D**. An in-house ensemble, fine-tuned on detonation data, returns all four in **milliseconds instead of an hour**, fast enough to score every candidate. Two folds of roughly **84 M parameters** each. That speed is why it screens, and never decides.

Why train heads you will never use?

Some of the score model's heads, the small output layers stacked on one shared trunk, never steer anything; at sample time they are simply not invoked, though they stay in the shipped model to regularise the trunk. That sounds wasteful. It is the opposite.

THE PROBLEM

Only **~3,000** trustworthy labels. A network with one job on a small dataset memorises it.

THE FIX

Give the shared trunk **several related jobs at once**. Each extra task constrains the representation the others depend on.

shared trunk • ~8 M parameters

almost all of the model lives here

↓ ↓ ↓ ↓ ↓ ↓ ↓

Viability
STEERS

Sensitivity
STEERS

Hazard
STEERS

Performance
TRAINS ONLY

SA
TRAINS ONLY

SC
TRAINS ONLY

Performance

Predicts the four detonation numbers (p, D, P, HOF). Deliberately kept **off** the steering bus: property targets are pursued through conditioning instead.

SA

Synthetic Accessibility. A drug-domain score for how hard a molecule looks to make. Calibrated on pharmaceutical reaction corpora, so it penalises the polynitro chemistry we want; production sets its steering scale to zero.

SC

Synthetic Complexity. A learned complexity score from a reaction corpus. Never wired into sample-time steering at all.

This is **multi-task regularisation**. The extra jobs are not there for their own answers: they force the trunk to encode *general* chemistry rather than a shortcut that satisfies one objective. The heads we use then read from a better trunk.

Each head is roughly 256k parameters against a ~8 M trunk, so **what the six heads share is most of the model**. Once training ends the three auxiliary outputs are discarded. What survives is the trunk.

One trunk, six heads, trained together

WHY Mid-generation there is no finished molecule to inspect, only a noisy vector. We need a judge that can **READ THAT HALF-FORMED STATE** and still say something useful.

The score model judges a **noisy** latent, which matters, because at generation time everything is noisy. All six heads share a single trunk and are trained **jointly**, not separately.

4-block FiLM-MLP (FiLM: side information enters as a per-layer scale and shift), shared 1024-d trunk, ~8 M parameters, six heads at ~256k each.

It reads the latent directly and never decodes. In training it sees real molecules' latents noised to *every* level; at generation it meets the same kind of noisy latent, now arriving from pure noise instead of from a molecule. Same distribution, so the judge stays in-distribution exactly when it has to.

THREE HEADS STEER

Viability ↑ **Sensitivity** ↓ **Hazard** ↓

Each has an independent on/off switch at sample time.

THREE HEADS REGULARISE

Performance, SA, SC

Never used for steering. They exist to improve the shared representation.

Self-distillation: let the model generate, catch the molecules that game the surrogate, feed them back as hard negatives. Two rounds, 918 hard negatives, refining the viability head.

Labelling and steering are different jobs

The Random Forest and Uni-Mol never steer generation. They run **once, offline**, to manufacture training labels. Only the score model is in the sampling loop.

WHEN	ON WHAT DATA	WHO RUNS	HOW OFTEN
Labelling before training	the 66k cached corpus	Random Forest, Uni-Mol ensemble, SMARTS catalogue, Politzer–Murray fit	once, offline
Training	noised latents + those labels	score model learns to predict the labels; denoiser learns to generate	many epochs
Steering at sample time	the half-noised molecule being built	score model only	every one of 40 steps
Re-ranking after decoding	surviving candidates	Uni-Mol returns, to score finished molecules	once per candidate

Why it has to work this way: the Random Forest and Uni-Mol can only read a *finished* molecule, and mid-diffusion there is no molecule yet, only a noisy vector. Their knowledge is **transferred once into the score model**, which steers on their behalf.

The denoiser, and how it learns

WHY Everything so far only *describes* molecules. This is the part that **INVENTS** them.

This is the network that actually invents molecules: a **44.6 M parameter** FiLM-ResNet working entirely on 1024-number latents. Training it is a fill-in-the-blank exercise.

- 1 Take a **real** molecule from the corpus, as its cached latent z_0
- 2 Add a random amount of noise to it $z_t, t \in 1..1000$
- 3 Also hand it **the property targets** and **the mask** saying which of them count c, m
- 4 Ask it to predict **the noise that was added**. That is the whole loss ϵ -prediction

The properties enter through **FiLM**: the four numbers are turned into per-layer scale and shift factors applied inside the network, so the conditioning reaches every block rather than being concatenated once at the input.

The trick that makes both modes possible: 10 % of the time the whole mask is zeroed during training, so the same network also learns to denoise with *no* targets at all. One set of weights, two behaviours, which is exactly what classifier-free guidance, steering by the gap between a conditioned and an unconditioned run, needs at sample time.

1000-step variance-preserving DDPM (denoising diffusion probabilistic model) in latent space. Two are trained, DGLD-H and DGLD-P, giving the guided lanes their diversity.

The mask is where trust gating actually happens

WHY Two awkward facts have to enter training: most rows lack some properties, and the ones they have differ in trust. The mask handles **BOTH AT ONCE**.

At every training step, a four-bit switch decides which properties are allowed to guide this molecule.

ρ density 1

D velocity 0

P pressure 1

HOF 0

- 1 Pick how many properties to use 0–4
- 2 Pick which, weighted so **trusted labels are chosen more often** tier weights A 1.0, B 0.7, C 0.3, D 0.1: low-trust labels are down-weighted, never excluded tier-weighted
- 3 Randomly switch some off again $p = 0.30$
- 4 Sometimes zero the whole mask → **no steering at all** $p = 0.10$

An all-zero mask is the unconditional branch: nothing steers. This is where low-trust Tier C/D labels mostly contribute, teaching what a plausible molecule looks like rather than hitting a specific number. When they *are* picked as targets, their low tier weight keeps the vote quiet.

CAREFUL: TWO SEPARATE SWITCHES

The mask covers only the **four property targets** (ρ , D, P, HOF), which reach the denoiser as conditioning. The steering heads (viability, sensitivity, hazard) are **not** masked: they have their own scales, set once at sample time, and push on chemistry quality. The score model's performance head is deliberately left off the steering bus, so property targets are pursued through conditioning alone.

Noise in, molecule out: in 40 steps

WHY Training is finished. Now we ask the model for molecules it has never seen, and steer it while it works.

- 1 Start from **pure random noise** in latent space $z_T \sim N(0, I)$
- 2 Denoise step by step (a DDIM sampler does it in 40 passes, not 1000), with two forces pushing at each step 40 DDIM steps
- 3 Decode the final vector back to a molecule frozen decoder

GUIDANCE 1 • CLASSIFIER-FREE (CFG)

Steers toward the **four property targets** (ρ , D, P, HOF).

The denoiser was trained both with and without the targets, so at sample time we run it twice and amplify the difference. **No separate predictor is needed.**

scale $w = 7$

GUIDANCE 2 • CLASSIFIER (SCORE MODEL)

Steers on **chemistry quality**: viability up, sensitivity down, hazard down.

Uses a **separately trained** network that can read a half-noised molecule, and adds its gradient at every step. Each head has its own scale.

$s_{\text{viab}} 1.0 \cdot s_{\text{sens}} 0.3 \cdot s_{\text{haz}} 1.0$

Why both? They answer different questions. CFG asks “*hit these numbers*”; the score model asks “*be a sane, safe molecule*”. Targets alone would happily produce something that scores well and cannot exist.

What a “lane” is, and what gets merged

WHY A single run explores one region and stops. To cover more chemistry we run several, then combine what they found.

A **lane** is one complete sampling run with its settings fixed throughout: (which denoiser, which property targets, which guidance). Run it and you get a list of molecules.

→ draw noise → 40 guided steps → decode

one lane

H and P are not guidance signals; they are two denoisers. Both receive all four property targets and run at identical guidance settings. They differ only in how they were trained: each oversampled a different extreme, so DGLD-H reliably supplies high heat-of-formation leads and DGLD-P supplies high-density ones. Four lanes run with different settings, so each explores a different part of chemical space:

LANE	DENOISER	TRAINED TO FAVOUR	GUIDANCE	OUTPUT
1	DGLD-H 44.6 M	the heat-of-formation tail	guided	≥40,000
2	DGLD-P 44.6 M	the density / pressure tail	guided	≥40,000
3	DGLD-H	same	unguided	≥40,000
4	DGLD-P	same	unguided	≥40,000

What is merged: the molecule lists, not the models. Each lane finishes and produces plain SMILES text; the lists are pooled and de-duplicated by canonical SMILES, so a molecule found twice is kept once. That pool enters the filter funnel. Nothing is averaged, and no weights are combined.

What the other methods produce

All four were run by us, not quoted from their papers, and scored through the same surrogate audit. Two share our corpus and compute (SMILES-LSTM, SELFIES-GA); MolMIM and REINVENT 4 run out of domain, so the parity is in the scoring, not the training setup. The composite scores are not on one scale, which is why the honest comparison is on DFT-audited D, ρ, P.

WHY Generating a plausible molecule is easy. The question is whether anything survives contact with physics, and that is only visible when you run the same audit on everyone.

METHOD	APPROACH	WHAT IT PRODUCED
SMILES-LSTM	language model over SMILES	Novel-looking output, but 18.3 % of it is copied verbatim from the training set
SELFIES-GA	genetic search over SELFIES	75 of 100 top candidates are corpus rediscoveries; its best novel one scored 9.73 and collapsed to 6.28 under DFT
MolMIM 70 M	large drug-domain VAE	Genuinely novel, but off-target: $D = 7.70 \text{ km s}^{-1}$
REINVENT 4	reinforcement learning	Novel high-nitrogen heterocycles peaking at $D = 9.02$, never DFT-audited

Be precise about REINVENT 4: we are *not* saying it failed. Its outputs are genuinely novel (we checked: only 0.04 % memorised) and its 9.02 was never put through DFT, so we genuinely do not know whether it would have held up or collapsed the way SELFIES-GA’s did. That is why the claim is worded as “the only method *confirmed* at the DFT level”, not “the only method that works”. An untested candidate is an open question, not a defeated rival.

Each fails a different way: one **memorises**, one **rediscovers**, one is **novel but weak**, one is **untested**. DGLD is the only one whose novel candidates were carried through DFT and stayed on target.

Five independent novelty checks

WHY A molecule already sitting in a database is not a discovery. And a generative model that quietly reproduces its training set will look brilliant unless you check, which is exactly how two of the baselines fail.

TEST	AGAINST WHAT	RESULT
Tanimoto window fingerprint overlap, 1 is identical	the 65,980-row training set, during the funnel's Stage 1	every candidate forced into $[0.20, 0.55]$: not a copy, not alien
Nearest-neighbour	same training set, after the funnel	97 of 100 have no neighbour above Tanimoto 0.55. Lead L1 sits at 0.27
Augmented corpus	all 694,518 rows, including motif-grafted variants	zero exact matches, and zero within Tanimoto 0.70
Public database	PubChem	96 of 97 absent. The one hit is a known tetrazole at rank 56
Patent literature	SureChEMBL patent corpus	all eleven leads absent

Stated honestly: novelty is established against *these* sources. A full search of the CAS Registry, the commercial index of known compounds, through SciFinder is recommended before any formal novelty claim, and the paper says so rather than implying the search was exhaustive.

Can anyone actually make it?

WHY A molecule nobody can build is a drawing. But the tools that answer this question were built for pharmaceuticals, so a negative answer has to be read carefully.

Retrosynthesis is run with **AiZynthFinder**, which searches backwards from the target for a sequence of known reactions ending in purchasable starting materials. It uses public USPTO reaction templates, drawn from US patent filings, and the ZINC in-stock catalogue.

L1: ROUTE FOUND

9 productive routes. The best is **4 steps** of standard synthetic chemistry: ring nitration, Boc protection, then a DPPA-mediated Curtius rearrangement.

Hazard noted honestly: the acyl-azide intermediate is primary-explosive class and needs careful handling.

THE OTHER TEN: NONE FOUND

Zero productive routes within budget, and still zero at **5× budget** on L4 and L5.

The energetic intermediates come back `in_stock: false` against ZINC, which is a catalogue of drug-discovery reagents.

What the 1-of-11 rate does and does not mean. USPTO templates and the ZINC catalogue are both drug-domain resources; nitration and nitrolysis chemistry, the reactions that install nitro groups, is thinly represented in them. So this measures a **gap in the public template library**, not proof that ten molecules are unmakeable. The paper claims only L1 as synthesis-actionable and reports the rest as a scaffold-diversity result.

The four-stage funnel

WHY The model proposes; it does not verify. Everything it produces is a hypothesis until physics is applied.

Every candidate must survive all four. The first three are cheap and use learned or heuristic scores; the last is expensive and uses real physics.

STAGE	CHECK	REMOVES
1 • SMARTS gate	chemist rules; $SA \leq 5.0$; $SCScore \leq 3.5$; Tanimoto to training set in $[0.20, 0.55]$	junk, unmakeable, and near-copies of the training data
2 • Pareto rerank	scaffold-aware composite over the Uni-Mol ensemble, keeping what nothing else beats on every property at once	low predicted performers
3 • GFN2-xTB	fast electronic-stability triage with cheap approximate quantum chemistry (HOMO–LUMO gap)	molecules that will not hold together
4 • DFT audit	standard DFT recipes: B3LYP/6-31G(d) geometry + frequencies, then ω B97X-D3BJ/def2-TZVP energy	anything that only <i>looked</i> good

The Pareto reranker: ranking on five axes at once

WHY After the cheap gate, thousands of candidates remain. Before spending expensive physics we must rank them by overall quality, but quality is five objectives that trade against each other.

COMPOSITE SCORE S

One **0-to-1 number** that folds the five objectives, performance, viability, novelty, sensitivity and hazard alerts, into a single weighted rank. Easy to sort, but it averages the trade-offs away.

PARETO FRONT

The set of candidates that **nothing else beats on every objective at once**. Honest about the trade-offs, but it gives no single ordering.

DGLD uses both. The **Pareto layers** are the outer sort (how dominant a candidate is), and the **composite S** breaks ties within each layer. Keep the **top 100**. That composite S is the same number on the y-axis of the position plot.

THE COMPOSITE SCORE, EXACTLY

$$S = 0.45 S_{\text{perf}} + 0.20 S_{\text{viab}} + 0.15 S_{\text{novel}} + 0.20 (1 - S_{\text{sens}}) - 0.10 S_{\text{alerts}}$$

Each S_i is a 0-to-1 sub-score. Performance dominates (0.45); sensitivity enters as $(1 - S_{\text{sens}})$, so a **safer molecule scores higher**; hazard alerts are the only **penalty** (-0.10).

scaffold-aware · ~1,800 chem-pass → top 100 into GFN2-xTB

Novelty and synthesis are checked twice

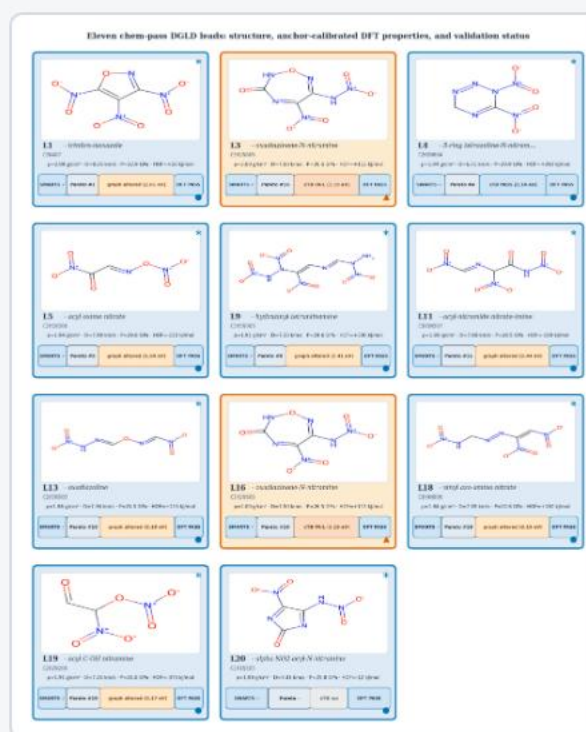
WHY A molecule that is already known, or that nobody can make, is not a discovery. Both have to be checked, and neither check is cheap enough to run on the whole pool.

QUESTION	CHEAP SCREEN, INSIDE STAGE 1	REAL CHECK, AFTER THE FUNNEL
Is it new?	Tanimoto to the nearest training molecule must sit in $[0.20, 0.55]$: not a copy, not alien	Search PubChem and the SureChEMBL patent corpus. 96 of 97 merged top-100 candidates are absent from PubChem
Can it be made?	Two quick complexity scores are capped: $SA \leq 5.0$ (Synthetic Accessibility , penalises rare fragments and awkward ring fusions) and $SCScore \leq 3.5$ (Synthetic Complexity , learned from a reaction database). Both are guesses from structure alone, no route is attempted.	AiZynthFinder actually searches for a route: working backwards from the target through known reactions until it reaches purchasable starting materials. L1 returns 9 productive routes , best is 4 steps; the other leads return none within budget

The funnel only uses **fast proxies**, because it has to run on 40,000 molecules. The database search and the retrosynthesis are far too slow for that, so they run afterwards on the handful that survived.

40,000 in, 11 out

raw sampled	40,000	100 %
after stages 1+2	~1,800	4.6 %
full filter	966	2.4 %
DFT-confirmed	11	10 unique



Paper Figure 2. The eleven chem-pass leads that survived the funnel.

The survivors are true local minima under DFT, with density and heat of formation computed from first principles and calibrated against six reference explosives.

A surrogate score is a hypothesis. DFT is the test.

A BASELINE THAT STOPPED EARLY

SELFIES-GA's best novel candidate scored 9.73 km s^{-1} on the surrogate.

Under DFT audit it fell to **6.28**.

THE HEADLINE LEAD, AUDITED

3,4,5-trinitro-1,2-isoxazole

ρ 2.09 g cm^{-3} · D 8.25 km s^{-1}

HMX 7.52 · PETN 8.24 (same recipe)

Tanimoto 0.27 to training set

Every number DGLD reports has been through the same audit that broke the baseline. That is the whole argument.